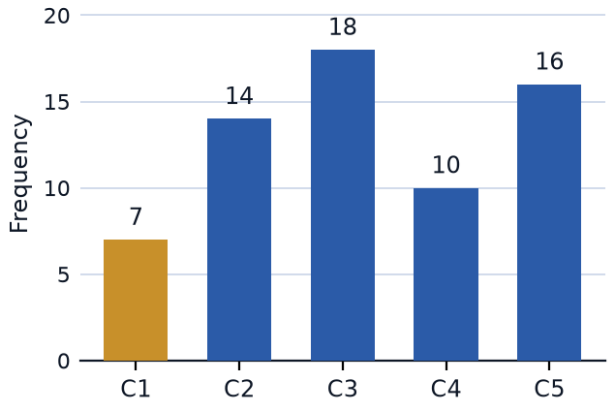
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The function $f(x) = a \cdot b^x$ passes through the points $(1, 12)$ and $(3, 108)$. Determine the value of a and the value of b, where $a > 0$ and $b > 0$.

2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The first four terms of an arithmetic sequence are $7, 11, 15, 19, \ldots$. Which expression represents the nth term, a_n, of the sequence for all integers $n \geq 1$?



3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

An arithmetic sequence is defined by its first term $a_1 = 7$ and its common difference d. The sum of the first n terms of this sequence is given by $S_n = \frac{n}{2}(2a_1 + (n-1)d)$. If the sum of the first 10 terms is $S_{10} = 115$, find the value of the common difference d.

4 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The first three terms of an arithmetic sequence are given by $a_1 = 5$, $a_2 = 8$, and $a_3 = 11$. What is the 10th term of this sequence?

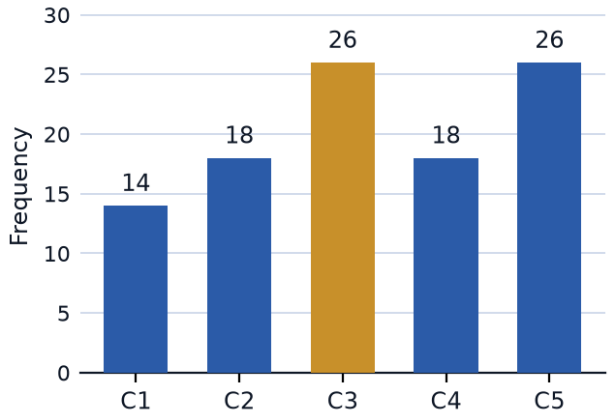
5 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Consider the arithmetic sequence where the first term is 5 and the common difference is 3. What is the 4th term of this sequence?

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6 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Simplify the expression: $\frac{2^5 \cdot 2^3}{2^4}$.



7 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

What is the solution to the equation $2^{x+1} = 32$?

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

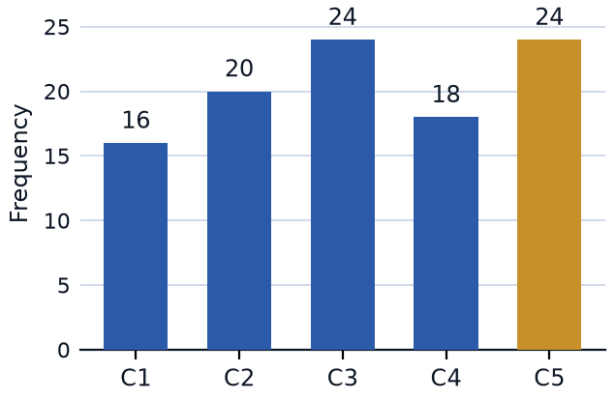
A set of five exam scores has a mean of 84. Four of the scores are 78, 86, 92, and 80. Find the 5th score, and find the new mean if the lowest score is removed.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A distribution of test scores has first quartile $Q_1 = 58$ and third quartile $Q_3 = 82$. Using the standard $1.5 \times \text{IQR}$ rule, what is the lowest score that would NOT be considered an outlier?

10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

[Statistics & Data Analysis · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.



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11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Level 3 · AdvancedIn the hard Module 2, questions 18–22 are typically high-difficulty multi-step problems with parameters k. If a student has 7 minutes remaining for the last 3 questions, which approach optimizes score yield: attempting all 3 partially, or using Desmos regression and choice elimination to guarantee complete accuracy on 2 of them?

12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

For what value of k will the system have no solution?

13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Solve the equation $\sqrt{2x+7} = x+2$ for all real values of x.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Evaluate the numerical value of $16^{-3/4} + \left(\frac{27}{8}\right)^{-2/3}$.

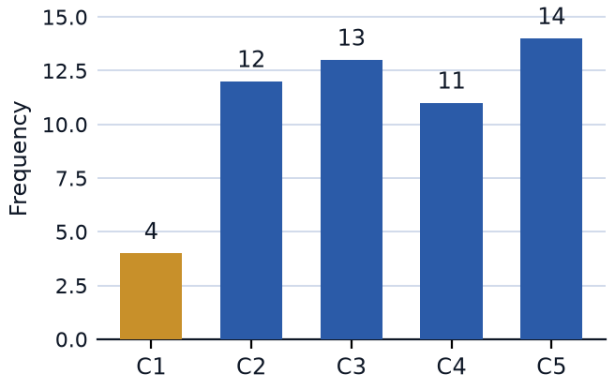
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 6.01 Systems & Inequalities

Find the sum of all integers between 10 and 100 (inclusive) that are divisible by 4.

16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 7.01 Sequences & Series

Find the equation of the line that passes through $(4,\ -1)$ and is perpendicular to the line $2x - 3y = 9$. Express your answer in slope-intercept form.



17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 8.01 Factoring Polynomials

A projectile is launched from ground level with height modelled by $h(t) = -16t^2 + 64t + 80$ feet after t seconds. Find (a) the maximum height reached, and (b) the time when it hits the ground.

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18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 9.01 Radicals & Exponents

In a geometric sequence, the 3rd term is 18 and the 6th term is 486. Find the common ratio r, the first term a_1, and the sum of the first 5 terms S_5.

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 1.01 Linear Equations & Lines

Solve the nonlinear system $y = x^2 - 3x + 2$ and $y = 2x - 4$ algebraically. State all intersection points (x, y).

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 10.01 Rational Expressions

Find the inverse function $f^{-1}(x)$ for $f(x) = \frac{3x+4}{2x-5}$, and state the domain of f^{-1}.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A data set recorded for review: operations with rational numbers consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean?

a. 16.5

b. 17.5

c. 18.5

d. 20.5

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A data set recorded for units resulting from a formula or graph consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean?

a. 19.5

b. 20.5

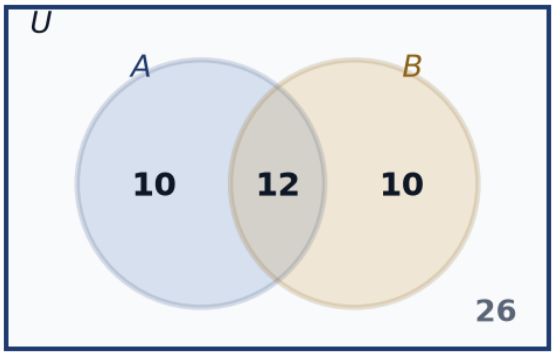
c. 21.5

d. 23.5

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23 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

The bar chart below shows the frequency distribution of an initial dataset of 10 test scores. Two additional scores, a and b, are added to this dataset to form a new dataset of 12 scores. The mean of the new dataset is 10, and its population variance is 26. What is the absolute difference $|a - b|$ between the two added scores? *(Note: The population variance of N values $y_1, y_2, \ldots, y_N$ with mean μ is $\sigma^2 = \frac{1}{N}\sum_{i=1}^{N}(y_i - \mu)^2$.)*



a. 8

b. 6

c. 10

d. 12

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A small company manufactures two types of artisanal cheeses: 'Mild Cheddar' and 'Sharp Provolone'. The production process involves two main stages: milk preparation and aging. The time (in hours) required for each kilogram of cheese at each stage is given in the table below. | Cheese Type | Milk Prep (hours/kg) | Aging (hours/kg) | |----------------|----------------------|------------------| | Mild Cheddar | 0.5 | 1.0 | | Sharp Provolone | 1.2 | 2.5 | Due to resource constraints, the company has a maximum of 120 hours available for milk preparation and 280 hours for aging per week. The profit per kilogram of Mild Cheddar is \x15. Let x be the number of kilograms of Mild Cheddar produced per week, and y be the number of kilograms of Sharp Provolone produced per week. Formulate a system of linear inequalities representing the production constraints and the objective function to maximize the total weekly profit. Which of the following options correctly represents the feasible region and the objective function?

a. Constraints: $0.5x + 1.2y \le 120$, $1.0x + 2.5y \le 280$, $x \ge 0$, $y \ge 0$. Objective function: $P = 8x + 15y$.

b. Constraints: $0.5x + 1.0y \le 120$, $1.2x + 2.5y \le 280$, $x \ge 0$, $y \ge 0$. Objective function: $P = 15x + 8y$.

c. Constraints: $0.5x + 1.2y \ge 120$, $1.0x + 2.5y \ge 280$, $x \ge 0$, $y \ge 0$. Objective function: $P = 8x + 15y$.

d. Constraints: $0.5x + 1.0y \le 120$, $1.2x + 2.5y \le 280$, $x \ge 0$, $y \ge 0$. Objective function: $P = 8x + 15y$.

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25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A student has scores of 81, 85, and 88 on three mathematics tests. What score must the student earn on the fourth test to have an average (arithmetic mean) score of exactly 87 across all four tests?

a. 87

b. 91

c. 94

d. 98

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

The table below shows the results of a survey on the preferred music genres of students in two different schools, School A and School B. The total number of students surveyed in School A is 200, and the total number of students surveyed in School B is 150. | Genre | School A | School B | |------------|----------|----------| | Pop | 70 | 55 | | Rock | 60 | 45 | | Electronic | 40 | 30 | | Classical | 30 | 20 | If a student is chosen at random from School A, what is the probability that they prefer Electronic music?

a. 0.20

b. 0.33

c. 0.25

d. 0.15

27 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A box plot shows Q1=20, Median=30, Q3=45. What is the IQR?

a. 15

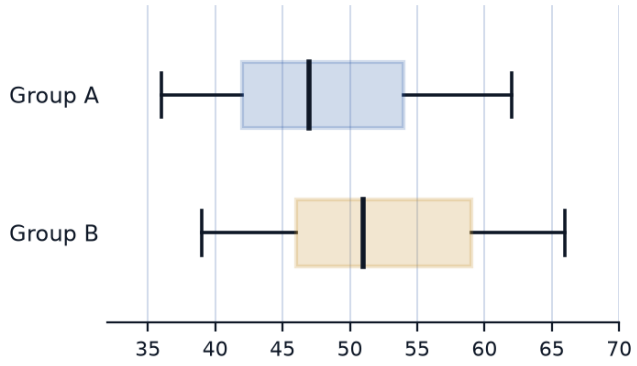
b. 25

c. 10

d. 45

28 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the median of the data set: $4, 6, 9, 20, 36, 40$.



a. 9

b. $\frac{31}{2}$

c. $\frac{29}{2}$

d. $\frac{27}{2}$

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

A data set has values $5, 7, 7, 8, 10$. What is the mean absolute deviation?

a. 1.5

b. 0.8

c. 1.2

d. 2.0

30 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

The standard deviation measures:

a. The most common value

b. Center of data

c. Spread of data around the mean

d. The range

31 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

Find the mean of the data set: $3, 8, 9, 12, 34$.

a. $\frac{71}{5}$

b. 9

c. $\frac{66}{5}$

d. $\frac{137}{10}$

32 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

A sample of 10 students scored: 70,75,80,80,85,85,85,90,90,100. What is the median?

a. 80

b. 84

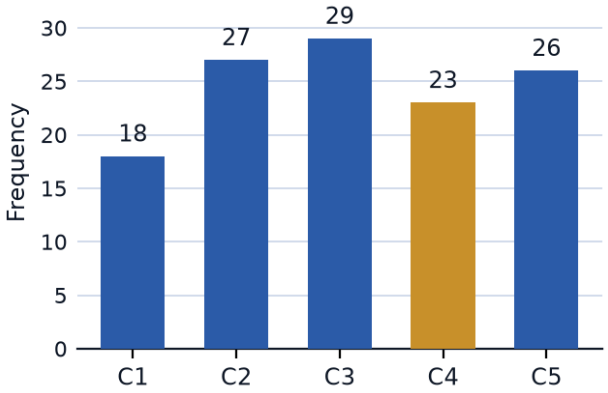
c. 85

d. 87.5

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33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A survey asks 50 students their favorite subject. This data is:



a. Ratio

b. Quantitative

c. Continuous

d. Categorical

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Which measure of center is most affected by outliers?

a. Median

b. Mean

c. Mode

d. Range

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A student scores 10 (1 times), 13 (1 times), 24 (1 times), 29 (3 times). What is the weighted average score?

a. $\frac{67}{3}$

b. $\frac{137}{6}$

c. $\frac{70}{3}$

d. $\frac{64}{3}$

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Which type of correlation does a scatter plot show if both variables increase together?

a. No correlation

b. Positive

c. Negative

d. Inverse

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37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

In a normal distribution, approximately what percent of data falls within 1 standard deviation of the mean?

a. 50%

b. 99%

c. 95%

d. 68%

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 9.01 Radicals & Exponents

Which measure is not affected by outliers?

a. Mean

b. Range

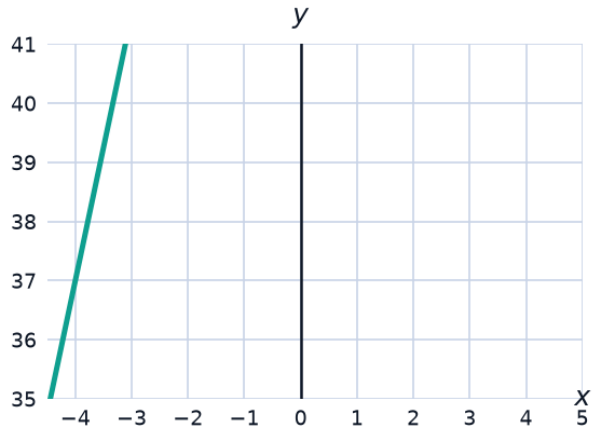
c. Median

d. Standard deviation

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39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 1.01 Linear Equations & Lines

A researcher is studying the relationship between the number of hours spent studying for a statistics exam and the score achieved. A random sample of 10 students yielded the following data, where x is the number of hours studied and y is the exam score: | Hours Studied (x) | Exam Score (y) | |---|---| | 2 | 65 | | 3 | 70 | | 4 | 78 | | 5 | 82 | | 6 | 88 | | 7 | 91 | | 8 | 95 | | 9 | 98 | | 10 | 100 | | 11 | 103 | The researcher models the relationship using a linear regression line of the form $y = \hat{a}x + \hat{b}$. Calculate the predicted exam score for a student who studies for $x = 7.5$ hours, given that the calculated regression line is $y = 4.5x + 55$.



a. 97.5

b. 95.0

c. 100.0

d. 92.5

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 10.01 Rational Expressions

Consider the following set of 11 exam scores listed in ascending order:

$37, 38, 40, 43, 45, 47, 47, 51, 51, 55, 56$ What is the interquartile range (IQR) of this data set?

a. 11

b. 14

c. 47

d. 19

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 2.01 Quadratic Functions

The table shows 5 students' exam scores. What is the average (arithmetic mean) of the data?

a. 82

b. 84

c. 86

d. 88

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42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 3.01 Exponents & Polynomials

Let $x, y,$ and z be real numbers satisfying the system of equations $\{x + y + z = 3; x^2 + y^2 + z^2 = 9$ What is the minimum possible value of $x^3 + y^3 + z^3$?

a. 15

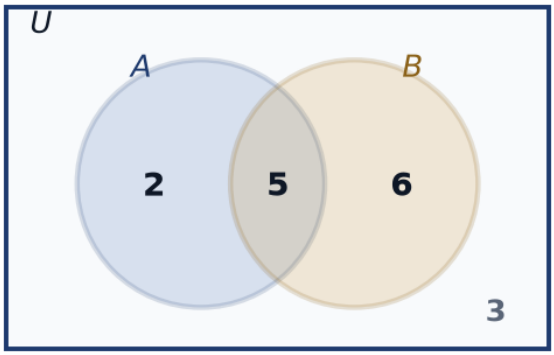
b. 21

c. 27

d. 33

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 4.01 Functions & Transformations

What is the complete solution set of the inequality $\log_{2x}(x^2 - 5x + 6) \leq 1$?



a. $(0, \frac{1}{2}) \cup [1, 2) \cup (3, 6]$

b. $[1, 2) \cup (3, 6]$

c. $(0, \frac{1}{2}) \cup [1, 6]$

d. $(0, \frac{1}{2}) \cup [1, 2] \cup [3, 6]$

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

In the figure, ΔABC is a right triangle with the right angle at vertex C, and altitude CD is drawn perpendicular to the hypotenuse AB. The lengths of the segments are given in terms of a variable x by $AD = x + 2$, $DB = 3x - 5$, and $CD = x + 5$. What is the perimeter of ΔABC?

a. 35

b. 50

c. 60

d. 150

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45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 6.01 Systems & Inequalities

Consider the system of inequalities: $\{x^2 + y^2 \leq 16; |x| + |y| \geq 4$ Which of the following statements best describes the region defined by this system? A. The region inside a circle of radius 4, excluding the interior of a square with vertices at $(\pm 4, 0)$ and $(0, \pm 4)$. B. The region inside a circle of radius 4, including the boundary of a square with vertices at $(\pm 4, 0)$ and $(0, \pm 4)$. C. The region outside a circle of radius 4, excluding the interior of a square with vertices at $(\pm 4, 0)$ and $(0, \pm 4)$. D. The region between a circle of radius 4 and a square with vertices at $(\pm 4, 0)$ and $(0, \pm 4)$.

a. A

b. B

c. C

d. D

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